1.8 Lebesgue Measure -- II

Continuity

Proposition 1.8.1: $A_1, \ldots, A_n \ldots$ are measurable and ascending, i.e. $A_n \subseteq A_{n+1}$. Then,

$$m(\bigcup_{n=1}^{\infty} A_n) = \lim_{n\to\infty} m(A_n)$$

. **$B_1, \ldots, B_n, \ldots$ are measurable and descending, i.e. $B_n \supseteq B_{n+1}$. Furthermore, $m(B_1) < \infty$. Then,**

$$m(\bigcap_{n=1}^{\infty} B_n) = \lim_{n\to\infty} m(B_n)$$

.

Proof: Let $E_1 = A_1$ and $E_n = A_n - A_{n-1}$ when $n \geq 2$. So

$$(1) \quad \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} E_n$$

.By countable additivity of Lebesgue measure (Proposition 1.7.13), we have

$$(2) \quad m(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n) = \lim_{n\to\infty} \sum_{k=1}^{n} m(E_k)$$

Note that by finite additivity (Lemma 1.7.6), we have

$$(3) \quad \sum_{k=1}^{n} m(E_k) = m(\bigcup_{k=1}^{n} E_k) = m(A_n)$$

, since $A_n = \bigcup_{k=1}^{n} E_k$. Combining (1)-(3), we obtain

$$m(\bigcup_{n=1}^{\infty} A_n) = \lim_{n\to\infty} m(A_n)$$

For descending continuity, let $F_n = B_1 - B_n$ when $n \geq 1$. Then F_n are measurable and ascending. By De Morgan's Law, we have

$$(4) \quad \bigcup_{n=1}^{\infty} F_n = B_1 - \bigcap_{n=1}^{\infty} B_n$$

. Thus,

$$(5) \quad m(\bigcup_{n=1}^{\infty} F_n) = m(B_1 - \bigcap_{n=1}^{\infty} B_n)$$

By the result of the first part,

$$(6) \quad m(\bigcup_{n=1}^{\infty} F_n) = \lim_{n\to\infty} m(F_n) = \lim_{n\to\infty} m(B_1 - B_n)$$

Combining (5) and (6) we get

$$(7) \quad m(B_1 - \bigcap_{n=1}^{\infty} B_n) = \lim_{n\to\infty} m(B_1 - B_n)$$

. Since $m(B_1) < \infty$, so is $m(\bigcap_{n=1}^{\infty} B_n)$ and $m(B_n)$. By (7) and excision property, we have

$$m(B_1) - m(\bigcap_{n=1}^{\infty} B_n) = m(B_1) - \lim_{n\to\infty} m(B_n)$$

, which easily implies that

$$m(\bigcap_{n=1}^{\infty} B_n) = \lim_{n\to\infty} m(B_n)$$

, since $m(B_1)$ is finite. **QED**

Outer and Inner Approximation of Lebesgue Measure

A measurable set can be approximated by

- an "outer" open set
- an "inner" closed set
- a **finite** collection of open intervals

Lemma 1.8.2 (Excision Property): If A is measurable and has a finite measure, then for any $B \supseteq A$, we have

$$m^*(B) - m^*(A) = m^*(B - A)$$

Proof: Since A is measurable and $A \subseteq B$, we have $m^*(B) = m^*(A) + m^*(B - A)$. Since $m^*(A) < \infty$, we have the desired result. **QED**

Proposition 1.8.3 (Outer Approximation): The statement that E is measurable is equivalent to the following statements:

- (a) For any $\epsilon > 0$, there is an open set O such that $E \subseteq O$ and $m^*(O - E) < \epsilon$
- (b) There is a G_δ set G such that $E \subseteq G$ and $m^*(G - E) = 0$

Proof: We denote by (M) the statement that E is measurable.

Proof of $(M) \Rightarrow (a)$:

Suppose $m^*(E) < \infty$. By the definition of outer measure, for any $\epsilon > 0$, there is an open set O such that $m^*(O) < m^*(E) + \epsilon$ (To be specific, there exists a collection of open intervals $\{I_k\}$) such that $\sum m^*(I_k) < m^*(E) + \epsilon$. Then let $O = \bigcup I_k$. Clearly $m^*(O) \leq \sum m^*(I_k)$). Since E is measurable and has a finite measure, by the excision property, we have

$$m^*(O - E) = m^*(O) - m^*(E) < \epsilon$$

Now suppose $m^*(E) = \infty$. It is not difficult to see that E is a union of a countable collection of disjoint and measurable sets whose measures are finite. (For example, take $E_n = E \cap [a_n, a_n + 1)$), where $a_n \in Z$ is an integer. Then E_n is measurable and has a finite measure and $m(E) = \sum m(E_n)$).

So we can write $m^*(E) = \sum_{n=1}^{\infty} m^*(E_n)$, where $m^*(E_n) < \infty$ and $E_1, \ldots, E_n, \ldots$ are disjoint and measurable.

By the above discussion, for any $\epsilon > 0$, there exists O_n such that

$$m^*(O_n - E_n) = m^*(O_n) - m^*(E_n) < \frac{\epsilon}{2^n}$$

. Take $O = \bigcup_{n=1}^{\infty} O_n$. Then it is not difficult to see that $O - E \subseteq \bigcup_{n=1}^{\infty}(O_n - E_n)$ (For $x \in O - E$, $x \in O_k$ for some k and $x \notin E_n$ for any n. Then x must be belong $O_k - E_k$)

So

$$m^*(O - E) \leq m^*(\bigcup_{n=1}^{\infty}(O_n - E_n)) \leq \sum_{n=1}^{\infty} m^*(O_n - E_n) \leq \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

. Therefore, property (a) holds for E when $m^*(E) = \infty$.

Proof of $(a) \Rightarrow (b)$:

For any n, since (a) holds for E, there exists an open set O_n such that $m^*(O_n - E) < \frac{1}{n}$. Take $G = \bigcap_{n=1}^{\infty} O_n$. Then G is a G_δ set. In addition, $E \subseteq G$ and $G - E \subseteq O_n - E$ for any n. So

$$m^*(G - E) \leq m^*(O_n - E) < \frac{1}{n}$$

. The above inequality holds for arbitrarily large n. Thus we must have $m^*(G - E) = 0$.

Proof of $(b) \Rightarrow (M)$:

If (b) holds for E, then the set $G - E$ is measurable since its outer measure is 0. Since the measurable sets form a $\sigma-$algebra, the set $\overline{G - E}$ is measurable. Thus, $E = \overline{G - E} \cap G$ is also measurable.

QED

Proposition 1.8.4 (Inner Approximation): The statement that E is measurable is equivalent to the following statements:

- (c) For any $\epsilon > 0$, there exists a closed set F_n such that $F_n \subseteq E$ and $m^*(E - F_n) < \epsilon$.
- (d) There exists a F_σ set F such that $F \subseteq E$ and $m^*(E - F) = 0$

Proof: We rely on the results of Proposition 1.8.3.

For $(M) \Rightarrow (c)$:

Since E is measurable, so is $R - E$. Then there exists an open set $O_n \supseteq R - E$ such that $m^*(O_n - (R - E)) < \epsilon$. Take $F_n = R - E_n$ and F_n is a closed set. Furthermore, $F_n \subseteq E$.

Furthermore, note that $E - F_n = O_n - (R - E)$. (Because $E - F_n = E - (R - O_n) = E \cap O_n = O_n - (R - E)$). Thus, we have

$$m^*(E - F_n) = m^*(O_n - (R - E)) < \epsilon$$

For $(c) \Rightarrow (d)$:

For any n, there exists F_n such that $F_n \subseteq E$ and $m^*(E - F_n) < \frac{1}{n}$. Take $F = \bigcup_{n=1}^{\infty} F_n$. Then F is a F_σ set and $F \subseteq E$. In addition, $E - F \subseteq E - F_n$ for any n. Thus, $m^*(E - F) \leq m^*(E - F_n) < \frac{1}{n}$ for arbitrarily large n. As a result, we must have $m^*(E - F) = 0$

For $(d) \Rightarrow (M)$:

Since $E - F$ has an outer measure 0, it is measurable. Thus, $E = (E - F) \cup F$ is measurable.

QED

Proposition 1.8.5 (Approximation by a Finite Collection of Open Intervals): If E is measurable with a finite measure, then for any $\epsilon > 0$, there exists a finite disjoint collection of open intervals $I_1, \ldots, I_n$ such that $O = \bigcup_{k=1} I_k$ and

$$m^*(E - O) + m^*(O - E) < \epsilon$$

Proof: The idea is take a "finite" section of its outer open set U.

By Proposition 1.8.3, since E is measurable, there exists an open set U such that $E \subseteq U$ and $m^*(U - E) < \frac{\epsilon}{2}$. Since $m^*(E) < \infty$, we have $m^*(U) < \infty$.

Note that U is countable and disjoint collection of open intervals, $U = \bigcup_{k=1}^{\infty} I_k$. Then,

$$\sum_{k=1}^{\infty} l(I_k) < \infty$$

. Take a n such that $\sum_{k=n+1}^{\infty} l(I_k) < \frac{\epsilon}{2}$. Let $O = \bigcup_{k=1}^{n} I_k$.

So $O - E \subseteq U - E$ and thus

$$(1) \quad m^*(O - E) \leq m^*(U - E) < \frac{\epsilon}{2}$$

On the other hand, $E - O \subseteq U - O$, and thus

$$(2) \quad m^*(E - O) \leq m^*(U - O) = m^*(\bigcup_{k=n+1}^{\infty} I_k) = \sum_{k=n+1}^{\infty} m^*(I_k) < \frac{\epsilon}{2}$$

. Combining (1) and (2) we obtain the result as desired.

QED

Borel-Cantelli Lemma

Proposition 1.8.6: $\{E_n\}_{n=1}^{\infty}$ is a countable collection of measurable sets. If $\sum_{n=1}^{\infty} m(E_n) < \infty$, then almost all $x \in R$ belongs to at most finitely many of the E_n's.

Proof: This is to say that the "limsup" of E_n has a measure 0. That is

$$(1) \quad m(\bigcap_{n=1}^{\infty}(\bigcup_{k=n}^{\infty} E_k)) = 0$$

. We often write

$$\limsup E_n = \bigcap_{n=1}^{\infty}(\bigcup_{k=n}^{\infty} E_k)$$

To prove (1), let $F_n = \bigcup_{k=n}^{\infty} E_k$. Then $F_1, \ldots, F_n, \ldots$ are descending and measurable. By the continuity of Lebesgue measure (Proposition 1.8.1), we have

$$m(\bigcap_{n=1}^{\infty} F_n) = \lim_{n\to\infty} m(F_n) = \lim_{n\to\infty} \sum_{k=n}^{\infty} m(E_k)$$

. Since $\sum_{k=1}^{\infty} m(E_k) < \infty$ converges, then $\sum_{k=n}^{\infty} m(E_k) \to 0$ as $n \to \infty$.

QED

Excercise

Q1. Show that if a set E has positive outer measure, then there is a bounded subset of E that also has positive outer measure.

Solution: If E is bounded, we are done. Assume E is unbounded and $m^*(E) > 0$. Suppose it is not the case, i.e. any bounded subset of E has an outer measure 0. Now let $A_n = (-n, n)$ for $n = 1, 2, \ldots$. Let $E_n = E \cap A_n$. Then $\bigcup_{n=1}^{\infty} E_n = E$. E_n is bounded subset of E, and by assumption $m^*(E_n) = 0$. Then $m^*(E) = m^*(\bigcup_{n=1}^{\infty} E_n) \leq \sum_{n=1}^{\infty} m^*(E_n) = 0$

, which is a contradiction. **QED**

Q2. Show that if E has finite measure and $\epsilon > 0$, then E is a disjoint union of a finite number of measurable sets, each of which has measure at most ϵ.

Solution: Suppose E is bounded by $(-M, M)$. Then let $n = \lceil \frac{2M}{\epsilon} \rceil$ and $\epsilon_0 = \frac{2M}{n}$. Note that $\epsilon_0 \leq \epsilon$. Let $A_k = [-M + (k-1) \cdot \epsilon_0, -M + k \cdot \epsilon_0), k = 1, 2, .., n$. Let $E_k = E \cap A_k$. Then $E = \bigcup_{k=1}^{n} E_k$ and $\{E_k\}_{k=1}^{n}$ is a disjoint collection of a finite number of measurable sets. Each $m(E_k) \leq \epsilon_0 \leq \epsilon$.

Suppose E is unbounded. Since E has finite measure, there exists a M such that $E = E_1 \cup E_0 \cup E_2$ where $E_0 = E \cap (-M, M)$, $E_1 = E \cap (-\infty, M]$ and $E_2 = E \cap [M, \infty)$. In addition, M can be sufficiently large so that $m(E_2) + m(E_3) < \epsilon$. Since E_0 is bounded, by a similar argument, E_0 is a disjoint union of n measurable sets, each of which has measure at most ϵ. Thus, E is the disjoint union of $n + 2$ measurable sets, each of which has measure at most ϵ.

QED

Q3. Show that a set E is measurable, if and only if, for any $\epsilon > 0$ there exist a closed set F and an open set O such that $F \subseteq E \subseteq O$ and $m^*(O - F) < \epsilon$.

Solution: Suppose E is a measurable. By outer approximation, there is an open set O such that $m^*(O - E) < \frac{\epsilon}{2}$. By inner approximation, there is a closed set F such that $m^*(E - F) < \frac{\epsilon}{2}$.

Now note that $O - F = (O - E) \cup (E - F)$, because

$$\begin{aligned}
(O - E) \cup (E - F) &= (O \cap \bar{E}) \cup (E \cap \bar{F}) \\
&= [(O \cap \bar{E}) \cup E)] \cap [(O \cap \bar{E}) \cup \bar{F}] \\
&= [(O \cup E) \cap (\bar{E} \cup E)] \cap [(O \cup \bar{F}) \cap (\bar{E} \cup \bar{F})] \\
&= O \cap [R \cap \bar{F}], \quad since \;\; O \cup \bar{F} = R \\
&= O \cap \bar{F} \\
&= O - F
\end{aligned}$$

. Thus,

$$m^*(O - F) = m^*((O - E) \cup (E - F)) \leq m^*(O - E) + m^*(E - F) < \epsilon$$

. **QED**

Q4. Show that continuity of measure together with finite additivity of measure implies countable additivity of measure.

Solution: Let m be a measure function that possesses the continuity and finite additivity properties. Let $E_1, \ldots, E_n, \ldots$ be a countable disjoint collection of measurable sets. We would like to show

$$m(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} m(E_n)$$

To this end, let $A_n = \bigcup_{k=1}^{n} E_k$. Then $\{A_n\}$ is ascending and $\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} E_n$. By continuity of m, we have

$$m(\bigcup_{n=1}^{\infty} A_n) = \lim_{n \to \infty} m(A_n)$$

By finite additivity of m, we have

$$m(A_n) = \sum_{k=1}^{n} m(E_k)$$

. Therefore,

$$m(\bigcup_{n=1}^{\infty} E_n) = \lim_{n\to\infty} \sum_{k=1}^{n} m(E_k) = \sum_{n=1}^{\infty} m(E_n)$$

, as desired.

QED